THE QUESTION

Two names. A new possibility.

TRAINING EXAMPLES

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POSSIBLE OUTPUTS

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A small language model. Every number visible.

Predict chances. Then choose.

current: a

next	chance
a	0.00
n	0.25
v	0.25
END	0.50

choose
one token

The chosen token becomes the next context.

Give each name a beginning and an end.

START	a	n	n	a	END

START → how a name begins

END → when a name finishes

Character vocabulary: a n v

Four letters → five next-token predictions.

Turn adjacent pairs into counts.

START	a	n	n	a	END

NEXT TOKEN →

FROM ↓	a	n	v	END
START	0	0	0	0
a	0	0	0	0
n	0	0	0	0
v	0	0	0	0

0 / 9 observations

a → n and n → a are different observations.

Count 1 of 9: START → a

START	a	n	n	a	END

NEXT TOKEN →

FROM ↓	a	n	v	END
START	1	0	0	0
a	0	0	0	0
n	0	0	0	0
v	0	0	0	0

1 / 9
observations

+1

a → n and n → a are different observations.

Count 2 of 9: a → n

START	a	n	n	a	END

NEXT TOKEN →

FROM ↓	a	n	v	END
START	1	0	0	0
a	0	1	0	0
n	0	0	0	0
v	0	0	0	0

2 / 9 observations

+1

a → n and n → a are different observations.

Count 3 of 9: n → n

START	a	n	n	a	END

NEXT TOKEN →

FROM ↓	a	n	v	END
START	1	0	0	0
a	0	1	0	0
n	0	1	0	0
v	0	0	0	0

3 / 9 observations

+1

a → n and n → a are different observations.

Count 4 of 9: n → a

START | a | n | n | a | END

NEXT TOKEN →

FROM ↓	a	n	v	END
START	1	0	0	0
a	0	1	0	0
n	1	1	0	0
v	0	0	0	0

4 / 9 observations

+1

a → n and n → a are different observations.

Count 5 of 9: a → END

START	a	n	n	a	END

NEXT TOKEN →

FROM ↓	a	n	v	END
START	1	0	0	0
a	0	1	0	1
n	1	1	0	0
v	0	0	0	0

5 / 9
observations

+1

a → n and n → a are different observations.

Count 6 of 9: START → a

START	a	v	a	END

NEXT TOKEN →

FROM ↓	a	n	v	END
START	2	0	0	0
a	0	1	0	1
n	1	1	0	0
v	0	0	0	0

6 / 9 observations

+1

a → n and n → a are different observations.

Count 7 of 9: a → v

START	a	v	a	END

NEXT TOKEN →

FROM ↓	a	n	v	END
START	2	0	0	0
a	0	1	1	1
n	1	1	0	0
v	0	0	0	0

7 / 9 observations

+1

a → n and n → a are different observations.

Count 8 of 9: v → a

START	a	v	a	END

NEXT TOKEN →

FROM ↓	a	n	v	END
START	2	0	0	0
a	0	1	1	1
n	1	1	0	0
v	1	0	0	0

8 / 9
observations

+1

a → n and n → a are different observations.

Count 9 of 9: a → END

START | a | v | a | END

NEXT TOKEN →

FROM ↓	a	n	v	END
START	2	0	0	0
a	0	1	1	2
n	1	1	0	0
v	1	0	0	0

9 / 9 observations

+1

a → n and n → a are different observations.

What followed a?

n	v	END	END

	a	n	v	END
count	0	1	1	2

Four observations. END owns two of them.

Divide by the row total.

	a	n	v	END
count	0	1	1	2
divide	0/4	1/4	1/4	2/4
chance	0.00	0.25	0.25	0.50

$$P(\text{next} = n \mid \text{current} = a) = 0.25$$

Training is finished.

Token labels
Boundary rules
Probability table

NEXT TOKEN →

FROM ↓	a	n	v	END
START	1.00	0.00	0.00	0.00
a	0.00	0.25	0.25	0.50
n	0.50	0.50	0.00	0.00
v	1.00	0.00	0.00	0.00

Generation and evaluation use this fixed table.

Begin at START.

START

NEXT TOKEN →

FROM ↓	a	n	v	END
START	1.00	0.00	0.00	0.00
a	0.00	0.25	0.25	0.50
n	0.50	0.50	0.00	0.00
v	1.00	0.00	0.00	0.00

OUTPUT

—

No answer key. Each selection continues the sequence.

Choose a from the START row.

START a

NEXT TOKEN →

FROM ↓	a	n	v	END
START	1.00	0.00	0.00	0.00
a	0.00	0.25	0.25	0.50
n	0.50	0.50	0.00	0.00
v	1.00	0.00	0.00	0.00

OUTPUT

a

No answer key. Each selection continues the sequence.

Choose n from the a row.

START a n

NEXT TOKEN →

FROM ↓	a	n	v	END
START	1.00	0.00	0.00	0.00
a	0.00	0.25	0.25	0.50
n	0.50	0.50	0.00	0.00
v	1.00	0.00	0.00	0.00

OUTPUT

an

No answer key. Each selection continues the sequence.

Choose a from the n row.

START | a | n | a

NEXT TOKEN →

FROM ↓	a	n	v	END
START	1.00	0.00	0.00	0.00
a	0.00	0.25	0.25	0.50
n	0.50	0.50	0.00	0.00
v	1.00	0.00	0.00	0.00

OUTPUT

ana

No answer key. Each selection continues the sequence.

Choose END from the a row.

START | a | n | a | END

NEXT TOKEN →

FROM ↓	a	n	v	END
START	1.00	0.00	0.00	0.00
a	0.00	0.25	0.25	0.50
n	0.50	0.50	0.00	0.00
v	1.00	0.00	0.00	0.00

OUTPUT

ana

No answer key. Each selection continues the sequence.

Greedy takes the largest probability.

GREEDY

START | a | END

SAMPLING: ONE POSSIBLE PATH

START | a | n | a | END

Same probabilities. Different selection rule.

A nice sample is not a score.

ana

Did the model predict well,
or did we pick a nice example?

Score held-out text with the table frozen.

This time, the text supplies the targets.

TRAIN: anna, ava SCORE: ana

	current	target	P(target)
1	START	a	1.00
2	a	n	0.25
3	n	a	0.50
4	a	END	0.50

Evaluation measures the model. It does not update it.

Every step must happen.

START	a	n	a	END
1.00	0.25	0.50	0.50	

$$P(\text{ana} + \text{END}) = 1 \times \tfrac{1}{4} \times \tfrac{1}{2} \times \tfrac{1}{2}$$

Sequence probability = 1/16 = 0.0625

A one-in-sixteen path.

EXPECTED MATCHING ATTEMPTS

16	16	4	2	1
× 1	× ¼	× ½	× ½	

Expected counts, not guaranteed sampling results.

Long products become tiny.

$$0.1 \times 0.1 \times 0.1 \times \ldots$$

1,000 factors → 10^{-1000}

Too small for ordinary floating point.

Use logs to turn the product into a sum.

A logarithm asks for the exponent.

$$10^3 = 1000$$

$$\log_{10}(1000) = 3$$

ln uses base $e \approx 2.718$

$$\ln(a \times b) = \ln(a) + \ln(b)$$

Turn each probability into a log.

	transition	P	ln P
1	START→a	1.00	0.000
2	a→n	0.25	−1.386
3	n→a	0.50	−0.693
4	a→END	0.50	−0.693

$\ln(1/16) \approx -2.773$

Negate: unlikely targets cost more.

	transition	P	−ln P
1	START→a	1.00	0.000
2	a→n	0.25	1.386
3	n→a	0.50	0.693
4	a→END	0.50	0.693

Negative log-likelihood: lower penalty is better.

Average over all four predictions.

	transition	P	−ln P
1	START→a	1.00	0.000
2	a→n	0.25	1.386
3	n→a	0.50	0.693
4	a→END	0.50	0.693

$(0 + 1.386 + 0.693 + 0.693) / 4 \approx 0.693$ nats

Lower than what?

	model	uses	NLL
1	uniform	equal chances	1.386
2	unigram	frequency	1.158
3	bigram	last character	0.693

Unigram targets: a:4, n:2, v:1, END:2

Same held-out example. Same vocabulary. Same metric.

Known characters. An unseen pair.

START	a	v	n	a	END

	a	n	v	END
v →	1.00	0.00	0.00	0.00

$$P(\text{avna} + \text{END}) = 1 \times \tfrac{1}{4} \times 0 \times \tfrac{1}{2} \times \tfrac{1}{2} = 0$$

Zero probability means an infinite NLL penalty.

Add one to every allowed outcome.

	a	n	v	END
observed	1	0	0	0
add 1	+1	+1	+1	+1
new count	2	1	1	1

New total: 1 + 4 = 5

Normalize again. Probability moves.

	a	n	v	END
observed	1	0	0	0
add 1	+1	+1	+1	+1
new count	2	1	1	1
chance	0.40	0.20	0.20	0.20

P(n | v): 0 → 0.20   P(a | v): 1 → 0.40

One is a choice. We can add k.

$$\frac{\text{observed count} + k}{\text{row total} + 4 \times k}$$

k = 0: no smoothing   k = 1: add one

Add k to all four outcomes, then normalize.

A safer guess can score worse.

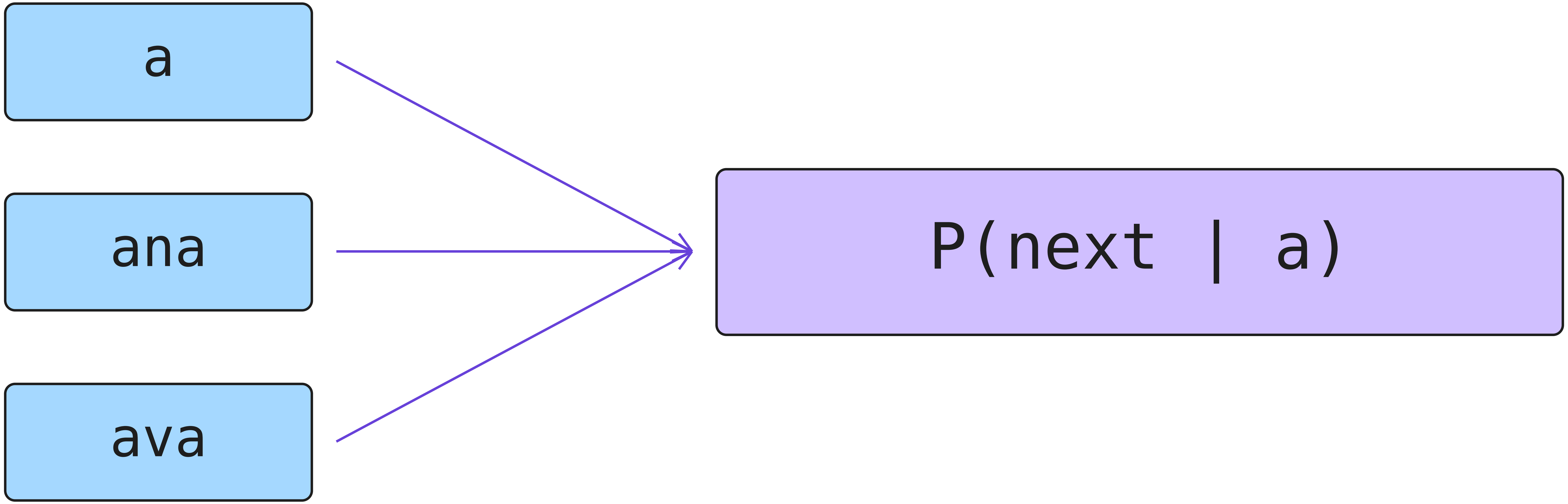
	model	NLL on ana
1	unsmoothed	0.693
2	add one	1.040

Smoothed across every row, including START.

Choose smoothing strength using validation data.

THE LIMITATION

Different histories. The same last character.



The model remembers one character, not the journey.

Every pair fits. The whole string can fail.

annnava

a→n n→n n→a a→v v→a

All observed. Still an awkward result.

Local plausibility is not global coherence.

WHAT COMES NEXT

First, make the code prove it.

NEXT VIDEO

1. Reproduce this exact table and score.
2. Run it on a larger names dataset.

THEN: TWO CHARACTERS OF MEMORY

Can more context fix what we just saw?

Count → normalize → generate → evaluate